Product Requirements Document (PRD)

Invoice Processing — From PDF to Decision | AI Solutions Associate Case Study | 2026

1. Executive Summary

Build a live, explainable invoice-processing workflow for a mid-size AP team. The system accepts a vendor invoice PDF, understands its content and layout, resolves the most likely purchase order (PO), validates commercial and arithmetic controls, checks duplicates and policy exceptions, and produces one of three outcomes: APPROVE, REVIEW, or REJECT. Every material decision must be traceable to extracted evidence, deterministic rules, and confidence signals.

The core design is deliberately hybrid. A multimodal model handles document understanding and ambiguous language/layout; deterministic software remains the authority for arithmetic, tolerances, duplicate detection, PO balances, currency/status rules, and the final decision. This follows the supplied invoice benchmark: native multimodal image processing substantially outperformed a Docling/structured-text approach on scanned receipts and invoices, while performance varied by model and document quality. It also addresses the AP study's finding that reconciliation and fraud detection were perceived as especially important risk-control mechanisms.

2. Problem Statement

- AP staff manually read heterogeneous invoice PDFs and compare them with POs, creating repetitive work and avoidable errors.
- Invoices may be text PDFs, scans, skewed images, low-quality documents, bundled line items, separate or embedded tax, missing identifiers, or inconsistent vendor formats.
- PO matching cannot safely be reduced to PO-number equality. References may be missing, vendors may split one PO across invoices, descriptions may differ, and amounts may be within an allowed tolerance.
- A black-box LLM decision is unsuitable for financial control. The system must show what it extracted, which PO candidates it considered, which rules passed/failed, and why the final decision was made.
- The case-study requirement is an actual runnable workflow with visible run stages, history, outputs, and realistic edge cases.

3. Target Users

User	Primary Need	Key Actions
AP Analyst	Process invoices quickly and safely	Upload/view invoice, inspect extraction, review PO match, resolve exceptions, approve/reject
AP Manager	Control exceptions and policy	Review escalations, inspect audit trail, override with reason
Procurement/Finance Admin	Maintain reference data	Load POs/vendors, configure tolerances and policies
Auditor/Controller	Trace decisions	View evidence, rules, source fields, timestamps, decision history

4. Product Goals

1. Automate the invoice-to-decision path from PDF ingestion through final disposition.
2. Maximize straight-through processing without sacrificing financial controls.

3. Use multimodal document understanding for messy PDFs/scans rather than making OCR the primary intelligence layer.
4. Make PO matching evidence-based, multi-signal, and tolerant of realistic business scenarios.
5. Separate probabilistic AI from deterministic financial policy.
6. Give AP staff a concise explanation that a non-technical user can audit.
7. Provide a live run view suitable for the case-study demo.

5. Non-Goals / Out of Scope for MVP

- Full ERP replacement or payment execution.
- Automatic bank transfer/payment initiation.
- Vendor self-service portal.
- Advanced tax-law determination across jurisdictions.
- Continuous model fine-tuning or training a custom VLM.
- Complex multi-entity accounting consolidation.
- Production-scale email ingestion with enterprise mail administration; a file upload is sufficient for the case-study MVP.

6. Core Workflow

8. Ingest invoice PDF and create a run ID.
9. Render PDF pages to images for multimodal analysis; preserve original PDF for audit.
10. Extract invoice header, vendor, dates, currency, totals, tax, PO references, payment details, and line items into a strict schema.
11. Run extraction verification/consistency checks; flag missing or contradictory fields but continue to PO resolution where useful.
12. Identify vendor using exact/fuzzy identifiers and reference data.
13. Generate PO candidates using deterministic filters plus fuzzy/semantic matching.
14. Score candidates using multiple independent signals rather than PO number alone.
15. Validate the selected candidate against PO status, vendor, currency, quantities, unit prices, tax, totals, tolerance, and remaining PO balance.
16. Run duplicate checks against historical invoices using vendor, invoice number, amount, date, hash/similarity, and PO context.
17. Apply deterministic decision policy: APPROVE, REVIEW, or REJECT.
18. Generate a human-readable reason and expose the complete evidence trail.
19. Persist run history and allow AP to inspect/resolve exceptions.

7. MVP Features

Priority	Feature	MVP Requirement
P0	PDF ingestion	Upload PDF; validate type/size; create run
P0	Multimodal extraction	Structured JSON with confidence/evidence per field
P0	Vendor resolution	Resolve vendor against seeded vendor master
P0	PO candidate retrieval	Return 0, 1, or multiple candidates
P0	PO matching	Multi-signal scoring with explicit match reasons
P0	Deterministic validation	Totals, tax, currency, vendor, status, quantities, tolerance, remaining balance
P0	Duplicate detection	Exact and near-duplicate invoice checks
P0	Decision engine	APPROVE / REVIEW / REJECT with

		reasons
P0	Run timeline	Show extraction → matching → validation → duplicate → decision
P0	Decision detail	Show invoice fields, matched PO, comparisons, failed checks
P0	Seed data	Realistic vendor/PO/invoice fixtures
P1	Manual review	Allow reviewer to select/override PO with mandatory reason
P1	History	Filter runs by status/vendor/date
P2	Email ingestion	Optional post-MVP

8. PO Matching Strategy

Matching is a two-stage process: candidate generation followed by candidate scoring. This avoids asking an LLM to search an entire PO database and prevents a single weak signal from controlling a financial decision.

Signal	**Example**	**Role**
Explicit PO number	PO-10482 appears on invoice	Very strong deterministic signal
Vendor identity	Legal name, email/domain, tax/vendor ID	Hard/strong filter
Currency	USD vs EUR	Hard validation unless policy allows conversion
Amount compatibility	Invoice total vs remaining PO value	Strong ranking + validation
Line-item overlap	SKU/item code/description	Strong ranking
Quantity compatibility	Invoice qty <= remaining PO qty	Strong validation
Date relationship	Invoice date relative to PO date	Supporting signal
Reference text	Contract/order/reference number	Supporting signal
Semantic similarity	'Office laptops' vs 'ThinkPad T14 units'	Ambiguity resolver
Historical vendor behavior	Known split-invoice pattern	Supporting signal only

A candidate should not be accepted solely because its semantic similarity is high. The policy engine requires minimum hard constraints and a confidence margin between the top candidate and runner-up. If two POs are plausible, the system routes to REVIEW rather than inventing certainty.

9. Decision Policy

Outcome	**Meaning**	**Typical Conditions**
APPROVE	Safe for straight-through processing	Required fields present; vendor/PO resolved; hard controls pass; within tolerance; no duplicate; confidence/margin above threshold
REVIEW	Human judgment required	Missing field; multiple plausible POs; ambiguous line mapping; unusual split invoice; tolerance exception; low extraction confidence; duplicate suspicion
REJECT	Policy violation or invalid document	Known duplicate; cancelled/closed PO; vendor blocked; impossible arithmetic; currency conflict; material overbilling outside tolerance; unsupported invoice

10. User Stories

• As an AP analyst, I can upload an invoice PDF and see the workflow execute stage by stage.

• As an AP analyst, I can see the extracted invoice fields and the evidence used to obtain them.

• As an AP analyst, I can see why a PO was selected and what alternative POs were considered.

• As an AP analyst, I can see quantity, unit price, subtotal, tax, total, and remaining-PO comparisons.

• As an AP analyst, I can see duplicate warnings before an invoice is approved.

• As an AP analyst, I can understand a REVIEW decision without reading model logs.

• As an AP manager, I can override a decision only with an explicit reason.

• As an auditor, I can inspect the immutable sequence of inputs, checks, outputs, and decision reasons.

11. Edge Cases Covered in MVP

- Missing PO number but unique vendor + amount/line evidence identifies one PO.
- PO number present but belongs to a different vendor.
- Multiple POs have the same vendor and similar amounts; top candidate is not sufficiently separated.
- Invoice total is within tolerance but line quantity exceeds remaining PO quantity.
- Invoice is a legitimate partial/split invoice against a PO.
- Several invoices legitimately consume one PO; remaining PO balance is tracked.
- Invoice is over tolerance.
- Duplicate invoice number from same vendor.
- Same invoice content re-submitted under a different filename.
- Tax is embedded or separated and arithmetic needs normalization.
- Currency mismatch between invoice and PO.
- Closed/cancelled PO.
- Missing invoice number/date/total.
- Scanned/low-quality invoice with uncertain fields.
- Bundled invoice line does not map one-to-one to PO lines.
- Negative/credit invoice or credit note is outside normal positive-invoice policy.

12. Success Metrics

Metric	MVP Target	Measurement
Field extraction exact-match	≥90% on seeded test set; track field-level accuracy	Ground truth comparison
PO match precision	≥95% on cases marked auto-match	Correct matched PO / auto-matched cases
Unsafe auto-approval rate	0% on defined critical-control violations	Critical violations incorrectly approved
Straight-through rate	≥60% on happy-path fixture set	Approved without manual intervention
Duplicate detection recall	≥95% on seeded duplicates	Duplicates detected / duplicates present
Decision explainability	100% of runs have structured reason codes	Runs with reason + evidence
Workflow reliability	≥95% successful end-to-end runs in demo test set	Completed runs / started runs
Median demo processing time	<30 seconds excluding provider latency where practical	Run timer

13. Assumptions

- PO and vendor data are supplied as CSV/JSON fixtures for the case study.
- A single invoice is processed per run.
- USD is the default currency unless explicitly represented otherwise.
- Tolerance is configurable and seeded at a simple policy level for the demo.
- The system does not execute payment.
- The model provider is accessed through a server-side API key and can be swapped through a provider abstraction.

14. Risks & Mitigations

Risk	Impact	Mitigation
VLM extraction error	Wrong financial data	Strict schema, confidence/evidence, second verification call, deterministic arithmetic
Wrong PO match	High financial risk	Candidate retrieval + multi-signal score + runner-up margin + review
LLM hallucination	Incorrect decision	LLM cannot directly approve/reject; policy engine is authoritative
Prompt/model variability	Inconsistent outputs	Structured output, low temperature where available, validation, provider abstraction
Duplicate miss	Double payment risk	Exact + fuzzy/content similarity + vendor/number/date/amount checks
Poor scan quality	Missing fields	Image preprocessing, field-level confidence, REVIEW routing
API outage/rate limit	Workflow failure	Clear error state; retry once with backoff; no silent fallback decision
Over-engineering	Missed case-study deadline	P0-only architecture; seeded data; no ERP/payment integration

15. Acceptance Criteria

20. Given a valid invoice PDF and matching PO, the system extracts required fields, finds the correct PO, passes validation, checks duplicates, and returns APPROVE with reasons.
21. Given an invoice with a missing PO number but a unique valid candidate, the system continues matching rather than immediately rejecting.
22. Given two plausible POs with insufficient confidence separation, the system returns REVIEW and shows both candidates.
23. Given a duplicate invoice, the system does not return APPROVE.
24. Given a material amount mismatch outside tolerance, the system does not return APPROVE.
25. Given a vendor/PO mismatch, the system does not return APPROVE.
26. Every run exposes stage status, inputs/outputs, rule results, final decision, and reason codes.
27. A reviewer can inspect the invoice and selected/alternative PO evidence without opening raw application logs.
28. The system survives malformed PDFs and model/API failures with a visible error state and no false financial decision.
29. A live demo can run the happy path and at least two non-trivial edge cases from a clean environment.

16. Source Basis

Source-derived benchmark insight: the supplied 2025 invoice benchmark reports that native multimodal image processing outperformed Docling-based structured parsing on scanned receipts and invoices, including 87.46% vs 47.00% on scanned receipts and 92.71% vs 64.03% on scanned invoices for the strongest reported model. The benchmark also highlights difficulty with unstructured alphanumeric fields such as IBANs. These findings motivate multimodal extraction plus deterministic validation rather than relying on text parsing alone.

Source-derived AP study insight: the supplied 2026 AP/AR study reports strong perceived relationships between AI-enabled reconciliation, fraud detection, AP/AR automation and operational risk mitigation, with reconciliation showing the strongest reported correlation and standardized regression effect. These are survey-based associations, not causal proof; they inform which controls deserve visibility in the product, not the product's statistical claims.